

SMART CAMPUS COMPANION

Testing Documentation

Multi-Platform Software Development for Real-World Problems (CSE)

Date: May 23, 2026

Platform: Flutter (Android) · Firebase · OpenStreetMap

1. Testing Overview

This document covers all testing activities for the Smart Campus Companion mobile application. Testing was conducted across three levels — unit testing, integration testing, and User Acceptance Testing (UAT) — following the SDLC requirements for the CSE Final Project.

Property	Value
Testing framework	Flutter Test (flutter_test package)
Platform tested	Android (physical device)
Test levels	Unit, Integration, UAT
Total test cases	30 test cases across all modules
Pass rate	27/30 passed (90%)
UAT participants	3 real students from ASTU CSE department

2. Unit Testing

Unit tests verify that individual functions, models, and business logic work correctly in isolation. All unit tests are located in the test/ folder of the Flutter project and run using the flutter test command.

2.1 Model Tests

UserModel Tests

ID	Test Case	Steps	Expected Result	Status
UT-01	UserModel.fromMap() parses correctly	Create UserModel from a valid Firestore map with all fields	Model fields match input map values	PASS
UT-02	UserModel.isStudent returns true for student role	Create UserModel with role='student', check isStudent	Returns true	PASS
UT-03	UserModel.isStaff returns true for staff role	Create UserModel with role='staff', check isStaff	Returns true	PASS
UT-04	UserModel.initials extracts correctly	Create user with fullName='Dawit Alemu', check initials	Returns 'DA'	PASS
UT-05	UserModel.initials handles single name	Create user with fullName='Yared', check initials	Returns 'Y'	PASS

Schedule Model Tests

ID	Test Case	Steps	Expected Result	Status
UT-06	ScheduleModel.isNow() returns true during class time	Set day=today, start=now-30min, end=now+30min, call isNow()	Returns true	PASS
UT-07	ScheduleModel.isNow() returns false outside class time	Set start and end in the past, call isNow()	Returns false	PASS
UT-08	ScheduleModel.displayRoom shows new room on change	Create schedule with status='room_changed', newRoom='Hall A'	displayRoom returns 'Hall A'	PASS
UT-09	ScheduleModel.isCancelled detects cancelled status	Create schedule with status='cancelled', check isCancelled	Returns true	PASS
UT-10	ScheduleModel.fromMap() handles missing fields	Create from map with missing optional fields	No exception, defaults applied	PASS

Announcement Model Tests

ID	Test Case	Steps	Expected Result	Status
UT-11	AnnouncementModel.timeAgo returns 'min ago' for recent post	Set publishedAt to 5 minutes ago, call timeAgo	Returns '5 min ago'	PASS
UT-12	AnnouncementModel.timeAgo returns 'Yesterday' for old post	Set publishedAt to yesterday, call timeAgo	Returns 'Yesterday'	PASS
UT-13	AnnouncementModel.categoryLabel returns correct label	Create with category='urgent', check categoryLabel	Returns 'Urgent'	PASS
UT-14	AnnouncementModel.isRead defaults to false	Create from map without is_read field	isRead is false	PASS

2.2 Service Tests

ID	Test Case	Steps	Expected Result	Status
UT-15	AuthService._handleAuthError maps wrong-password correctly	Pass FirebaseAuthException with code='wrong-password'	Returns 'Incorrect password.'	PASS
UT-16	AuthService._handleAuthError maps user-not-found correctly	Pass FirebaseAuthException with code='user-not-found'	Returns 'No account found with this email.'	PASS
UT-17	CacheService caches and retrieves schedule	Cache 3 schedules, clear memory, retrieve from cache	Returns same 3 schedules	PASS
UT-18	CacheService returns empty list	Call getCachedSchedule()	Returns empty	PASS

	when cache is empty	on fresh cache	list, no exception	PASS
UT-19	FirestoreService._getDayName returns correct day	Call with weekday=1 (Monday)	Returns 'monday'	
UT-20	FirestoreService._getDayName returns correct day for Sunday	Call with weekday=7 (Sunday)	Returns 'sunday'	PASS

3. Integration Testing

Integration tests verify that multiple components work correctly together — including screens, services, and Firebase. These tests run on a real Android device connected via USB using flutter drive.

3.1 Authentication Flow

ID	Test Case	Steps	Expected Result	Status
IT-01	Login with valid student credentials navigates to dashboard	Enter valid email + password, tap login	Dashboard screen loads with student name and schedule	PASS
IT-02	Login with invalid credentials shows error message	Enter wrong password, tap login	Red error banner appears with message	PASS
IT-03	Staff login navigates to admin dashboard	Login with staff credentials	Admin dashboard loads with staff name and post buttons	PASS
IT-04	Logout returns user to login screen	Tap avatar, tap Sign out	Login screen displayed, user session cleared	PASS

3.2 Dashboard Integration

ID	Test Case	Steps	Expected Result	Status
IT-05	Dashboard loads today's schedule from Firestore	Login as student, open dashboard	Today's classes appear with correct room and time	PASS
IT-06	Dashboard shows announcements from Firestore	Post announcement in Firebase, reload dashboard	Announcement appears in recent announcements section	PASS

IT-07	Dashboard shows offline banner when disconnected	Disable device WiFi, open dashboard	Green offline banner appears at top	PASS
IT-08	Dashboard loads cached schedule when offline	Cache schedule, disconnect, reload	Cached classes show correctly without network	PASS
IT-09	Pull to refresh reloads data from Firestore	Swipe down on dashboard	Loading indicator shows, data refreshes	PASS

3.3 Announcements Integration

ID	Test Case	Steps	Expected Result	Status
IT-10	Announcements feed loads from Firestore in real time	Post new announcement in Firebase console	Appears in feed without manual refresh	PASS
IT-11	Urgent filter shows only urgent announcements	Tap Urgent chip with mixed announcements	Only urgent category items shown	PASS
IT-12	Search filters announcements by title	Type 'room change' in search bar	Only matching announcements shown	PASS
IT-13	Tapping announcement marks it as read	Tap unread announcement (green dot)	Green dot disappears, detail sheet opens	PASS
IT-14	Mark all read clears all unread dots	Tap Mark all read with 3 unread items	All green dots disappear, count resets to 0	PASS

3.4 Staff Admin Integration

ID	Test Case	Steps	Expected Result	Status
IT-15	Staff post creates announcement in Firestore	Login as staff, post announcement, check Firebase	Document created in announcements collection	PASS
IT-16	Posted announcement appears in student feed	Post from staff, check student account feed	Announcement visible immediately	PASS
IT-17	Delete removes announcement from Firestore	Delete post, confirm dialog, check Firebase	Document removed from announcements collection	PASS
IT-18	Urgent toggle makes post appear	Post with is_urgent=true,	Post appears	PASS

	at feed top	check student feed	pinned at top of student feed	
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4. User Acceptance Testing (UAT)

UAT was conducted with 3 real students from ASTU CSE department who were not involved in the development process. Each participant was given a test device with the app installed and asked to complete a set of tasks without assistance. Their feedback was recorded.

4.1 UAT Participants

Participant	Profile
Participant 1	3rd year CSE student, male, daily smartphone user
Participant 2	3rd year CSE student, female, daily smartphone user
Participant 3	2nd year CSE student, male, moderate smartphone user

4.2 UAT Task Results

ID	Test Case	Steps	Expected Result	Status
UAT-01	Login to the app as a student	Open app, select Student, enter credentials, tap login	Dashboard loads with correct name and schedule	PASS
UAT-02	Find today's class schedule	Navigate to dashboard, read today's schedule	All 3 participants found schedule in under 10 seconds	PASS
UAT-03	Find a specific announcement	Open announcements, use search to find 'room change'	All 3 found it — 2 used search, 1 scrolled	PASS
UAT-04	Identify an urgent announcement	Look for any urgent alert in the feed	All 3 identified the red banner correctly	PASS
UAT-05	Find Room 204 on the campus map	Open map, search 'Room 204', tap result	2/3 found it easily; 1 needed help with search	PASS
UAT-06	Find Dr. Haile Tadesse's contact info	Open directory, search 'Haile', open profile	All 3 completed in under 20 seconds	PASS
UAT-07	Navigate between screens using bottom nav	Tap each icon in bottom navigation bar	All 3 navigated correctly without instruction	PASS

UAT-08	Use the app while offline (WiFi disabled)	Disable WiFi, check dashboard	2/3 saw cached schedule correctly; 1 saw empty state (bug)	FAIL
UAT-09	Identify what changed when a room changes	Dashboard with room_changed status displayed	All 3 noticed the red text and 'changed' label	PASS
UAT-10	Log out of the app	Tap avatar icon, tap Sign out	All 3 returned to login screen successfully	PASS

4.3 UAT Feedback Summary

Category	Finding
Most liked feature	Real-time announcements and the urgent alert banner — all 3 participants said this was the most useful feature compared to WhatsApp groups
Most confusing part	The campus map search — one participant didn't know to type in the search bar and expected to tap the map directly
Suggested improvement	Add a notification badge count number on the bell icon (not just a dot)
Bug found	One participant's offline cache didn't load — cached schedule showed empty instead of last synced data (race condition in initState)
Overall rating	8.3/10 average across all 3 participants

4.4 Bugs Found During Testing

ID	Test Case	Steps	Expected Result	Status
BUG-01	Offline cache race condition	Disable WiFi before app fully loads, check dashboard	Empty schedule shown instead of cached data	FAIL
BUG-02	Search bar on map needs explicit tap	Open map, try typing without tapping search first	Keyboard doesn't appear unless bar is tapped first	PASS
BUG-03	Long staff names truncate in directory list	Staff member with full double-barrelled name	Name truncates at tile — full name visible in profile	PASS

4.5 Bug Fix — BUG-01 Offline Cache Race Condition

The offline cache bug was identified during UAT and fixed before final submission. The root cause was that the connectivity check in dashboard initState() ran asynchronously and sometimes completed after the Firestore request had already started, causing the cache branch to be skipped.

Fix applied: wrapped the full data loading logic in a try-catch that falls back to cache on any exception, regardless of connectivity status. This ensures cache is always used as a fallback when Firestore fails for any reason.

5. Test Summary

Metric	Result
Total unit tests	20 test cases
Unit tests passed	20/20 (100%)
Total integration tests	18 test cases
Integration tests passed	18/18 (100%)
Total UAT tasks	10 tasks
UAT tasks passed	9/10 (90%)
Bugs found	3 bugs (1 fixed, 2 minor non-critical)
Overall pass rate	47/48 test cases passed (98%)

All critical user flows — login, schedule viewing, announcements, map navigation, staff directory, and admin posting — passed both integration testing and UAT. The one failing UAT task (offline cache) was identified, root-caused, and fixed before final submission.

End of Testing Documentation